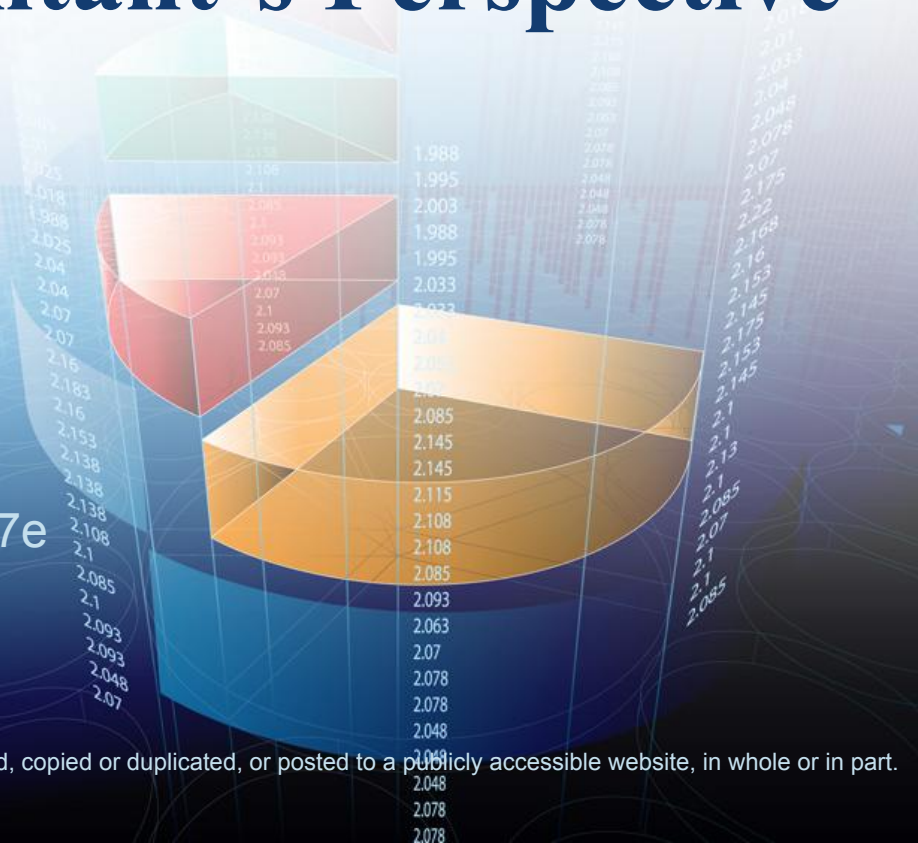
Chapter 1
The Information System: An Accountant's Perspective



Accounting Information Systems, 7e
James A. Hall

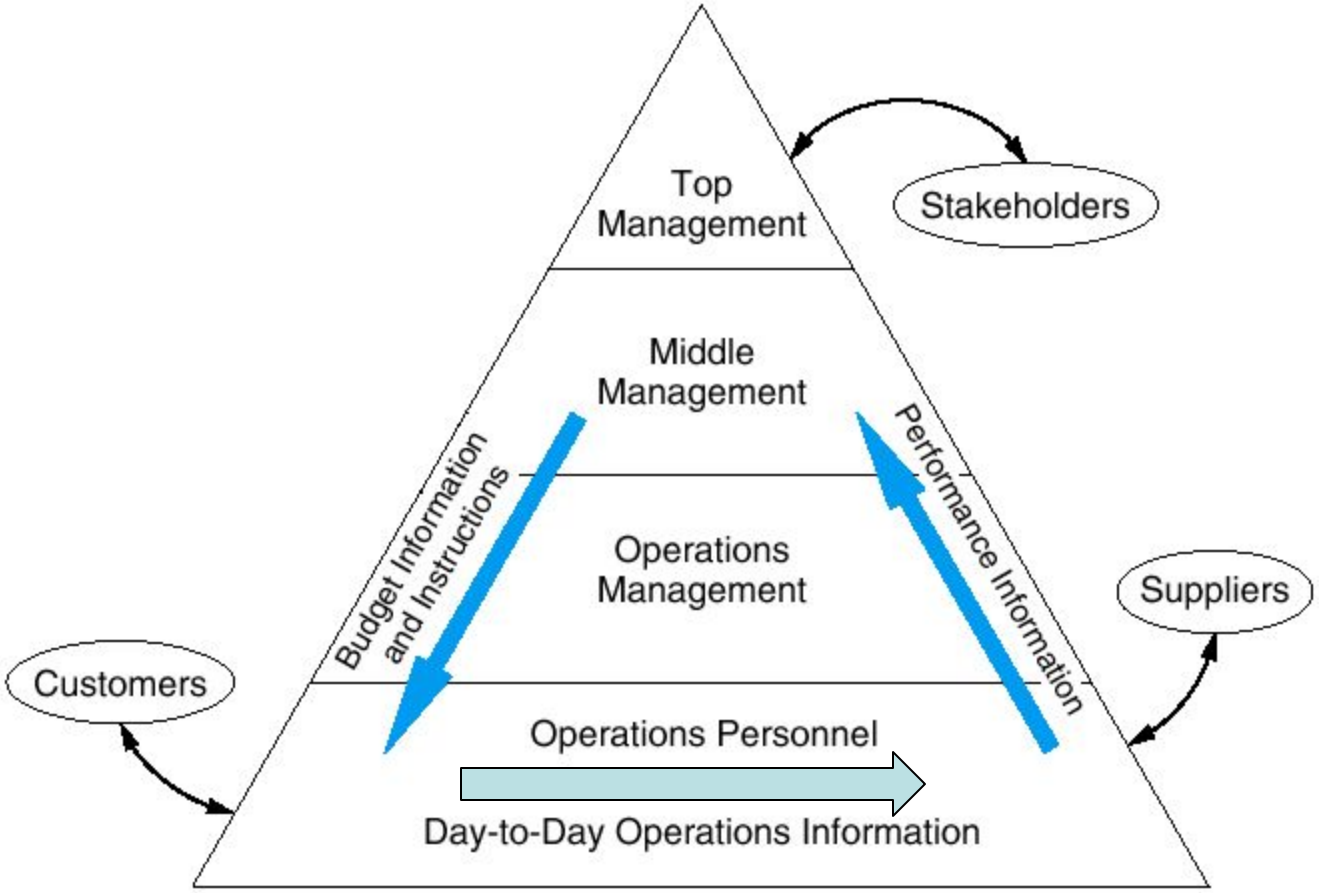
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Objectives for Chapter 1

- Understand the primary information flows within the business environment.
- Understand the difference between accounting information systems and management information systems.
- Understand the difference between Financial transactions and non-financial transactions.
- Know the general model for information systems.
- Be familiar with the functional areas of a business.
- Understand the stages in the evolution of information systems.
- Understand the relationship between the three roles of accountants in an information system.

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Internal & External Information Flows



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Internal Information Flows

- ***Horizontal*** flows of information used primarily at the operations level to capture transaction and operations data
- ***Vertical*** flows of information
  - downward flows — instructions, quotas, and budgets
  - upward flows — aggregated transaction and operations data

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Information Requirements

- Each user group has unique information requirements.
- The higher the level of the organization, the greater the need for more aggregated information and less need for detail.

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Information in Business

- Information is a ***business resource*** that***:***
  - needs to be appropriately managed
  - is vital to the survival of contemporary businesses

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What is a System?

- A group of interrelated multiple components or subsystems that serve a common purpose
- System or subsystem?
  - A **system** is called a subsystem when it is viewed as a component of a larger system.
  - A **subsystem** is considered a system when it is the focus of attention.

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System Decomposition versus System Interdependency

- **System Decomposition**
  - the process of dividing the system into smaller subsystem parts
- **System Interdependency**
  - distinct parts are not self-contained
  - they are reliant upon the functioning of the other parts of the system
  - all distinct parts must be functioning or the system will fail

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What is an Information System?

An information system is the set of formal procedures by which data are collected, processed into information, and distributed to users.

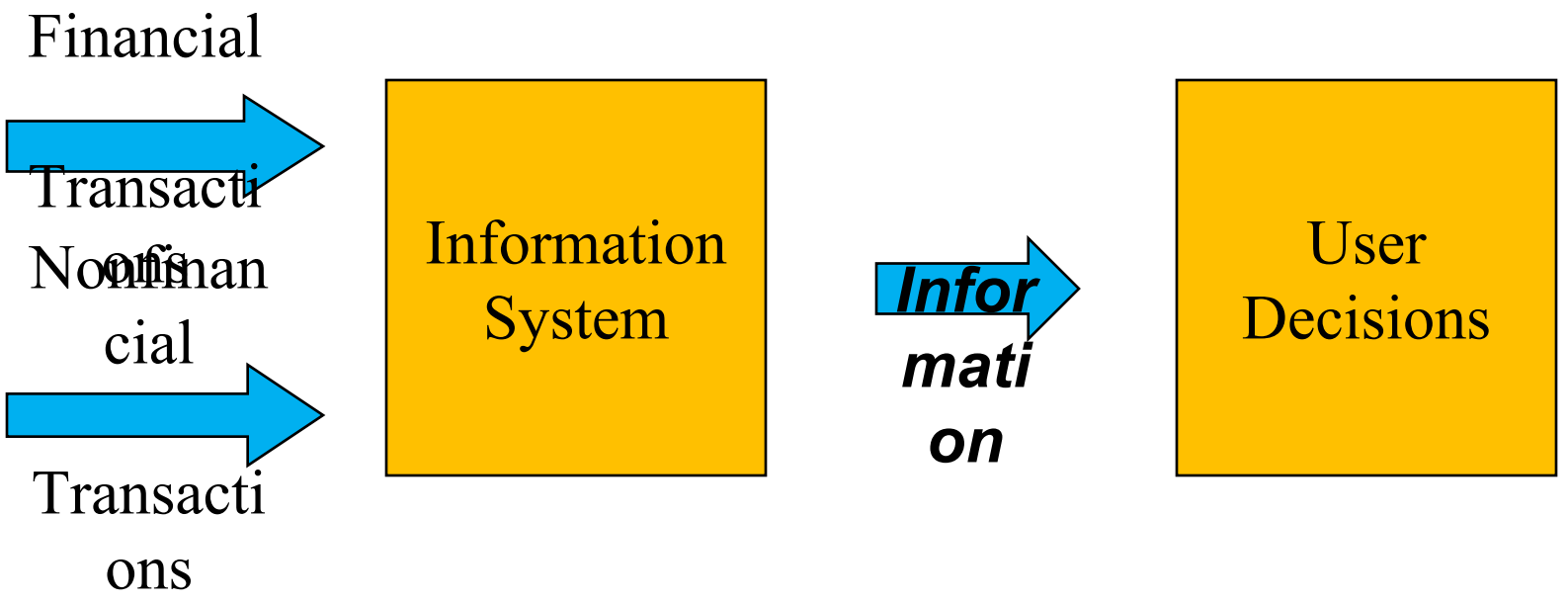
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Transactions

- A **transaction** is a business event**.**
- **Financial transactions**
  - economic events that affect the assets and equities of the organization
  - e.g., purchase of an airline ticket
- **Nonfinancial transactions**
  - all *other* events processed by the organization's information system
  - e.g., an airline reservation — no commitment by the customer

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Transactions



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What is an Accounting Information System?

- Accounting *is* an information system.
  - It identifies, collects, processes, and communicates economic information about a firm using a wide variety of technologies.
  - It captures and records the financial effects of the firm's transactions.
  - It distributes transaction information to operations personnel to coordinate many key tasks.

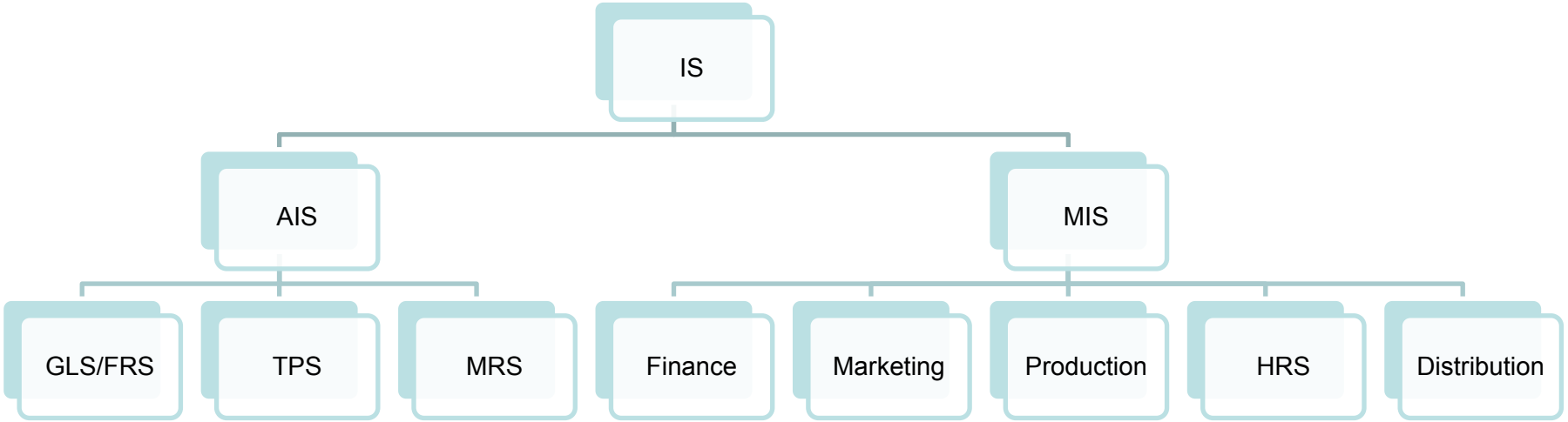
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AIS versus MIS

- **Accounting Information Systems** (AIS) process
  - financial transactions; e.g., sale of goods
  - nonfinancial transactions that directly affect the processing of financial transactions; e.g., addition of newly approved vendors
- **Management Information Systems** (MIS) process
  - nonfinancial transactions that are not normally processed by traditional AIS; e.g., tracking customer complaints

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AIS versus MIS?



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AIS Subsystems

- **Transaction processing system (TPS)**
  - supports daily business operations
- **General Ledger/ Financial Reporting System (GL/FRS)**
  - produces financial statements and reports
- **Management Reporting System (MRS)**
  - produces special-purpose reports for internal use

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General Model for AIS

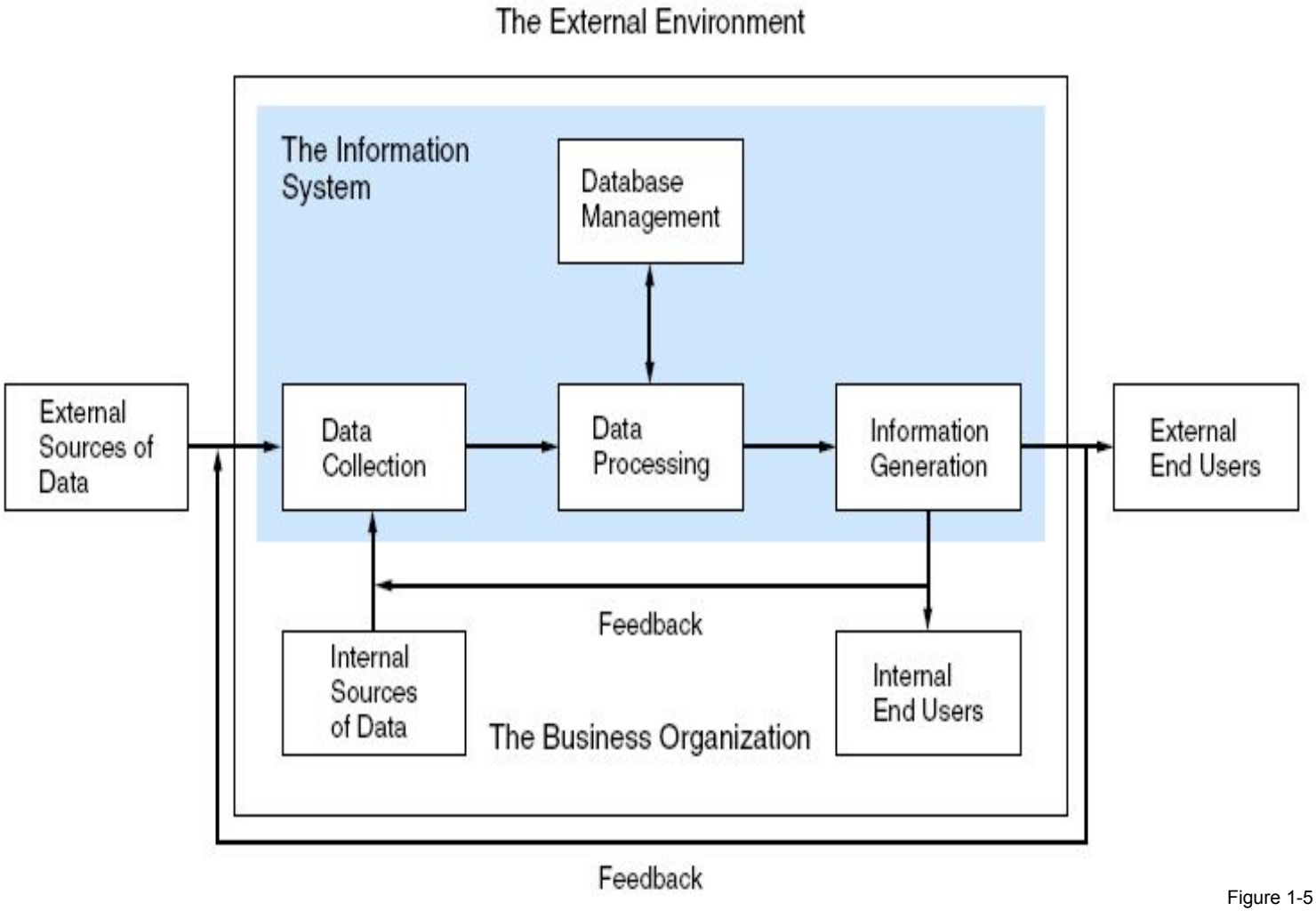


Figure 1-5

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Data Sources

- **Data sources** are financial transactions that enter the information system from internal and external sources.
  - *External financial transactions* are the most common source of data for most organizations.
    - E.g., sale of goods and services, purchase of inventory, receipt of cash, and disbursement of cash (including payroll)
  - *Internal financial transactions* involve the exchange or movement of resources within the organization.
    - E.g., movement of raw materials into work-in-process (WIP), application of labor and overhead to WIP, transfer of WIP into finished goods inventory, and depreciation of equipment

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Transforming the Data into Information

Functions for transforming data into information according to the general AIS model:

1. Data Collection
2. Data Processing
3. Data Management
4. Information Generation

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1. Data Collection

- Capturing transaction data
- Recording data onto forms
- Validating and editing the data

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2. Data Processing

- Classifying
- Transcribing
- Sorting
- Batching
- Merging
- Calculating
- Summarizing
- Comparing

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3. Data Management

- Storing
- Retrieving
- Deleting

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4. Information Generation

- Compiling
- Arranging
- Formatting
- Presenting

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Characteristics of Useful Information

- Regardless of physical form or technology, *useful* information has the following characteristics:
  - ***Relevance:*** serves a purpose
  - ***Timeliness:*** no older than the time period of the action it supports
  - ***Accuracy:*** free from material errors
  - ***Completeness:*** all information essential to a decision or task is present
  - ***Summarization:*** aggregated in accordance with the user's needs

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Information System Objectives in a Business Context

- The goal of an information system is to support
  - the stewardship function of management
  - management decision making
  - the firm's day-to-day operations

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Organizational Structure

- The structure of an organization helps to allocate
  - responsibility
  - authority
  - accountability
- **Segmenting** by business function is a very common method of organizing.

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Functional Areas

- Inventory/Materials Management
  - purchasing, receiving and stores
- Production
  - production planning, quality control, and maintenance
- Marketing
- Distribution
- Personnel
- Finance
- Accounting
- Computer Services

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Accounting Independence

- Information reliability requires **accounting independence.**
  - Accounting activities must be separate and independent of the functional areas maintaining resources.
  - Accounting supports these functions with information but does not actively participate.
  - Decisions makers in these functions require that such vital information be supplied by an independent source to ensure its integrity.

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The Computer Services Function

Distributed Data Processing

Most companies fall in between.

Centralized Data Processing

Reorganizing the computer services function into small information processing units that are distributed to end users and placed under their control

All data processing is performed by one or more large computers housed at a central site that serves users throughout the organization.

Primary areas:
database administration
data processing
systems development
systems maintenance

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Organization of IT Function in a Centralized System

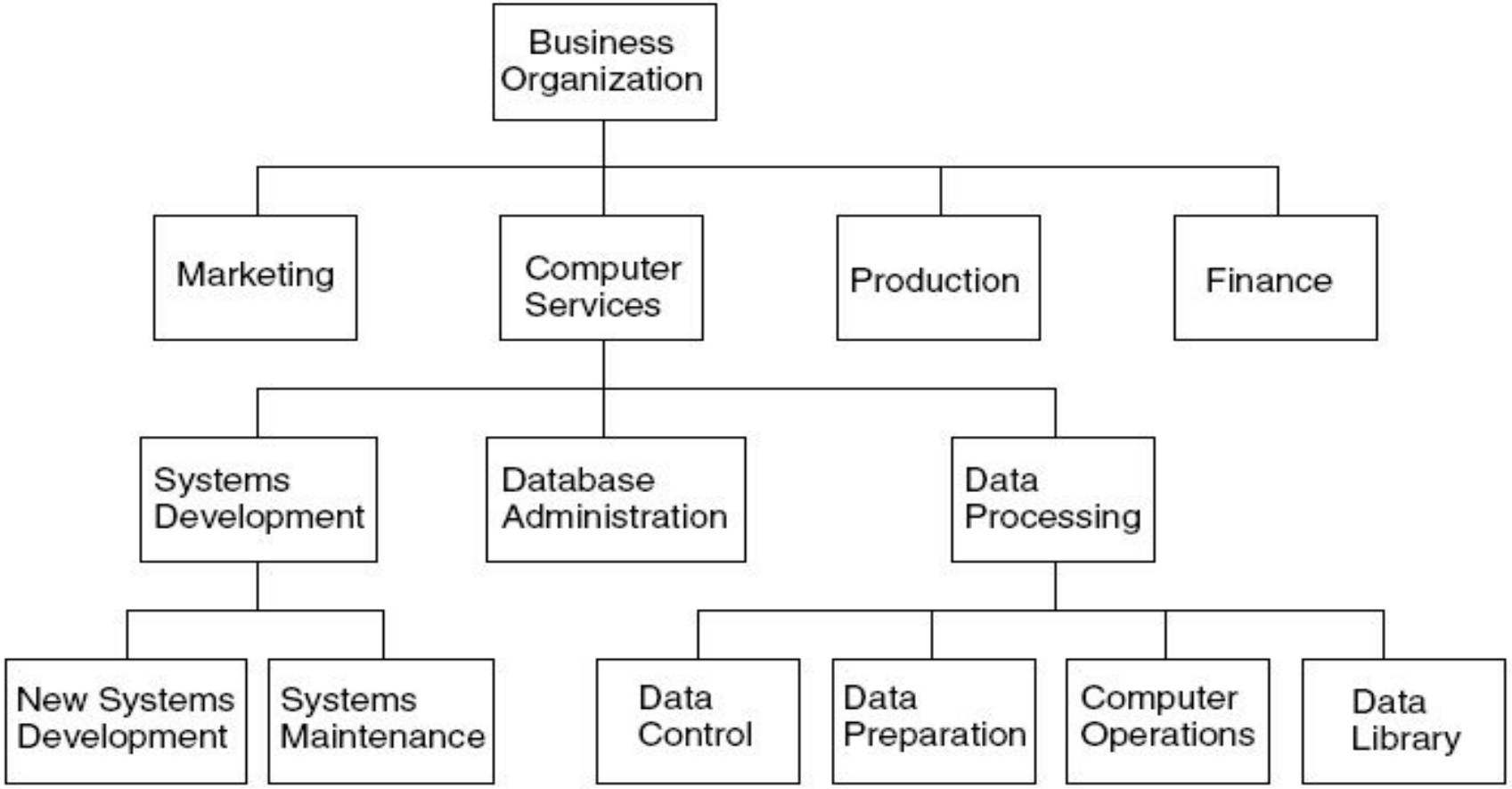


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Organizational Structure for a Distributed Processing System

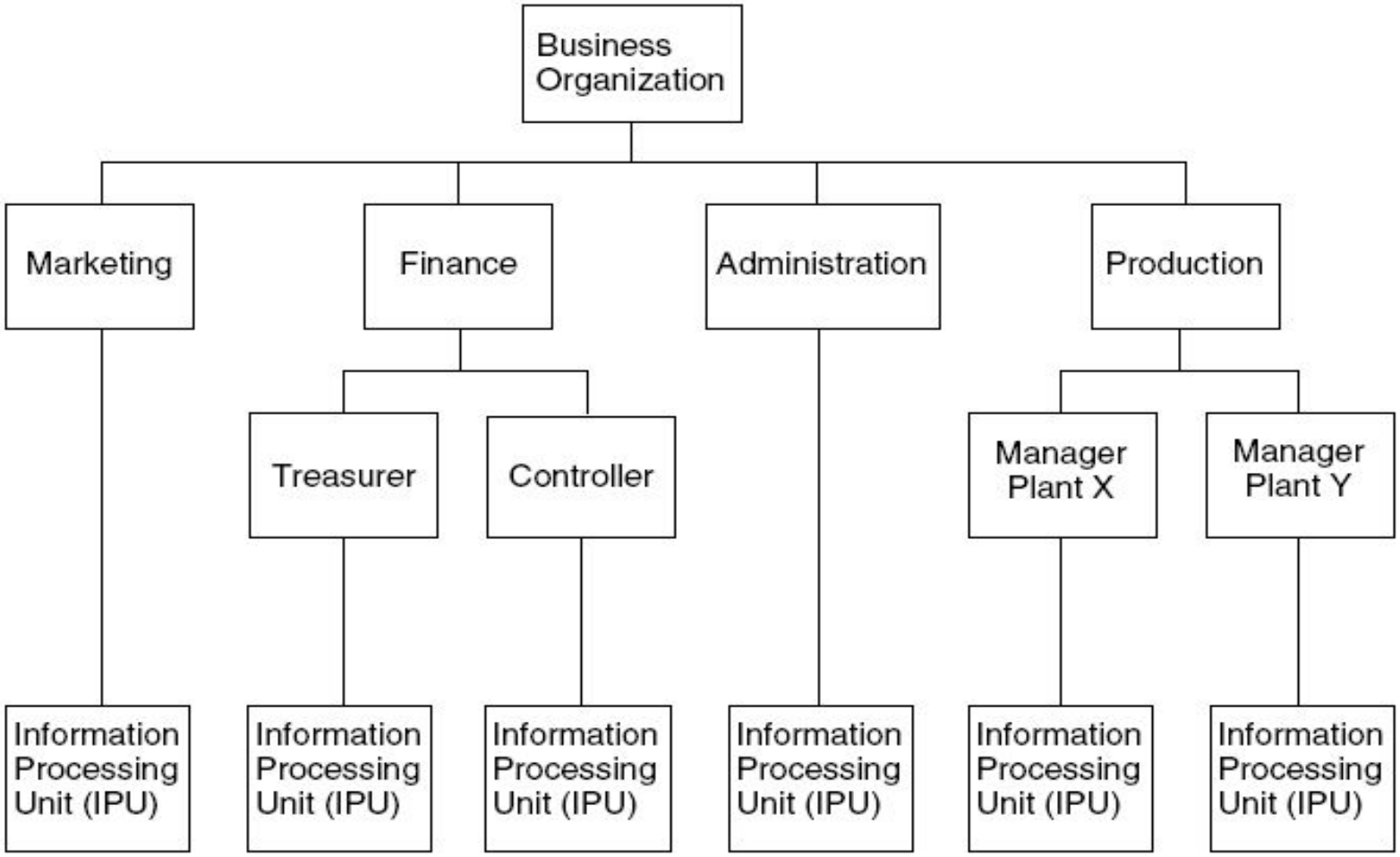


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Potential Advantages of DDP

- Cost reductions in hardware and data entry tasks
- Improved cost control responsibility
- Improved user satisfaction since control is closer to the user level
- Backup of data can be improved through the use of multiple data storage sites

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Potential Disadvantages of DDP

- Loss of control
- Mismanagement of company resources
- Hardware and software incompatibility
- Redundant tasks and data
- Consolidating tasks usually segregated
- Difficulty attracting qualified personnel
- Lack of standards

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Manual Process Model

- Transaction processing, information processing, and accounting are physically performed by people, usually using paper documents.
- Useful to study because:
  - helps link AIS courses to other accounting courses
  - often easier to understand business processes when not shrouded in technology
  - facilitates understanding internal controls

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The Evolution of IS Models: The Flat-File Model

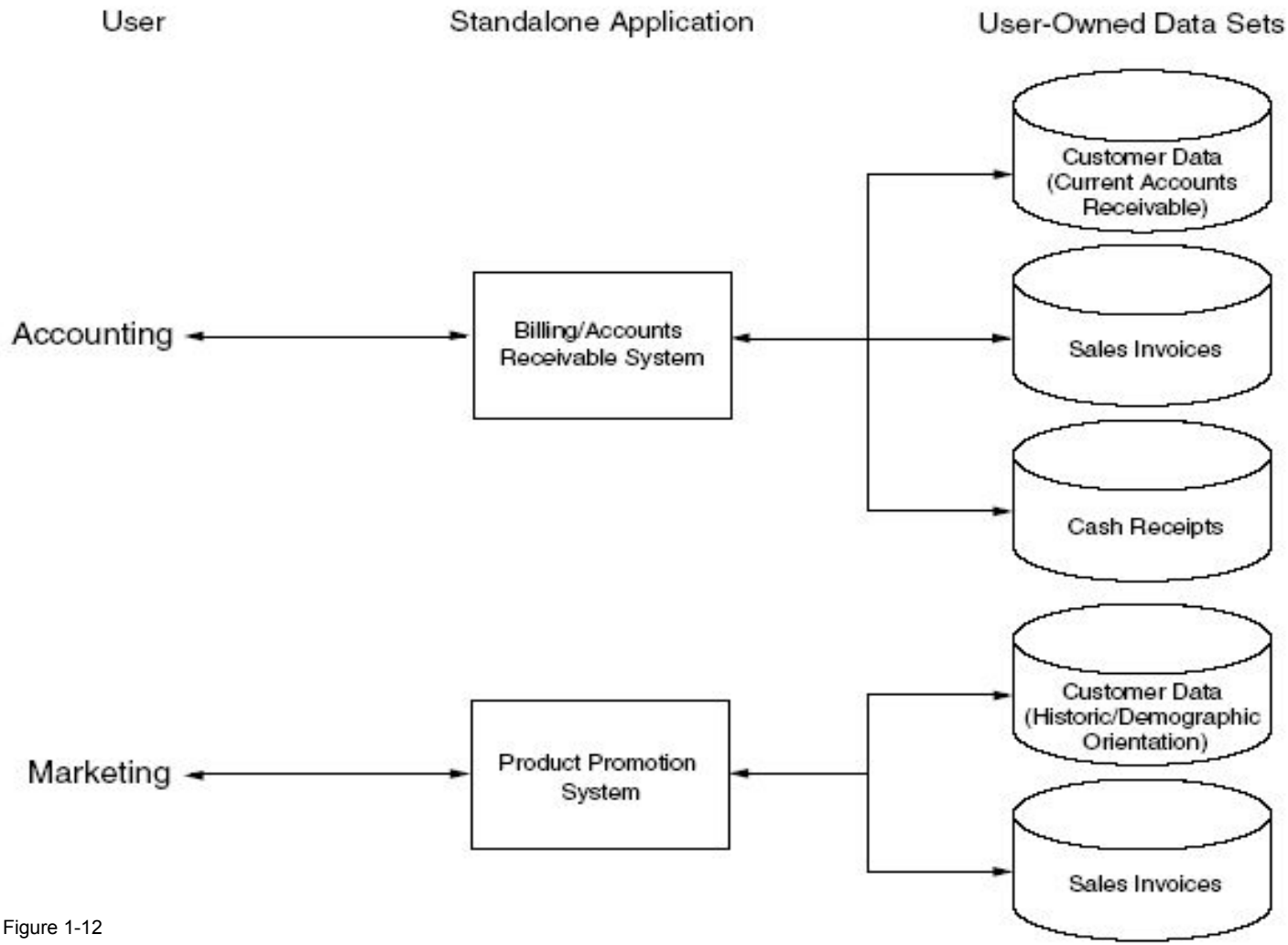


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Data Redundancy Problems

- **Data Storage** - excessive storage costs of paper documents and/or magnetic form
- **Data Updating** - changes or additions must be performed multiple times
- **Currency of Information** - potential problem of failing to update all affected files
- **Task-Data Dependency** - user's inability to obtain additional information as needs change
- **Data Integration** - separate files are difficult to integrate across multiple users

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The Evolution of IS Models: The Database Model

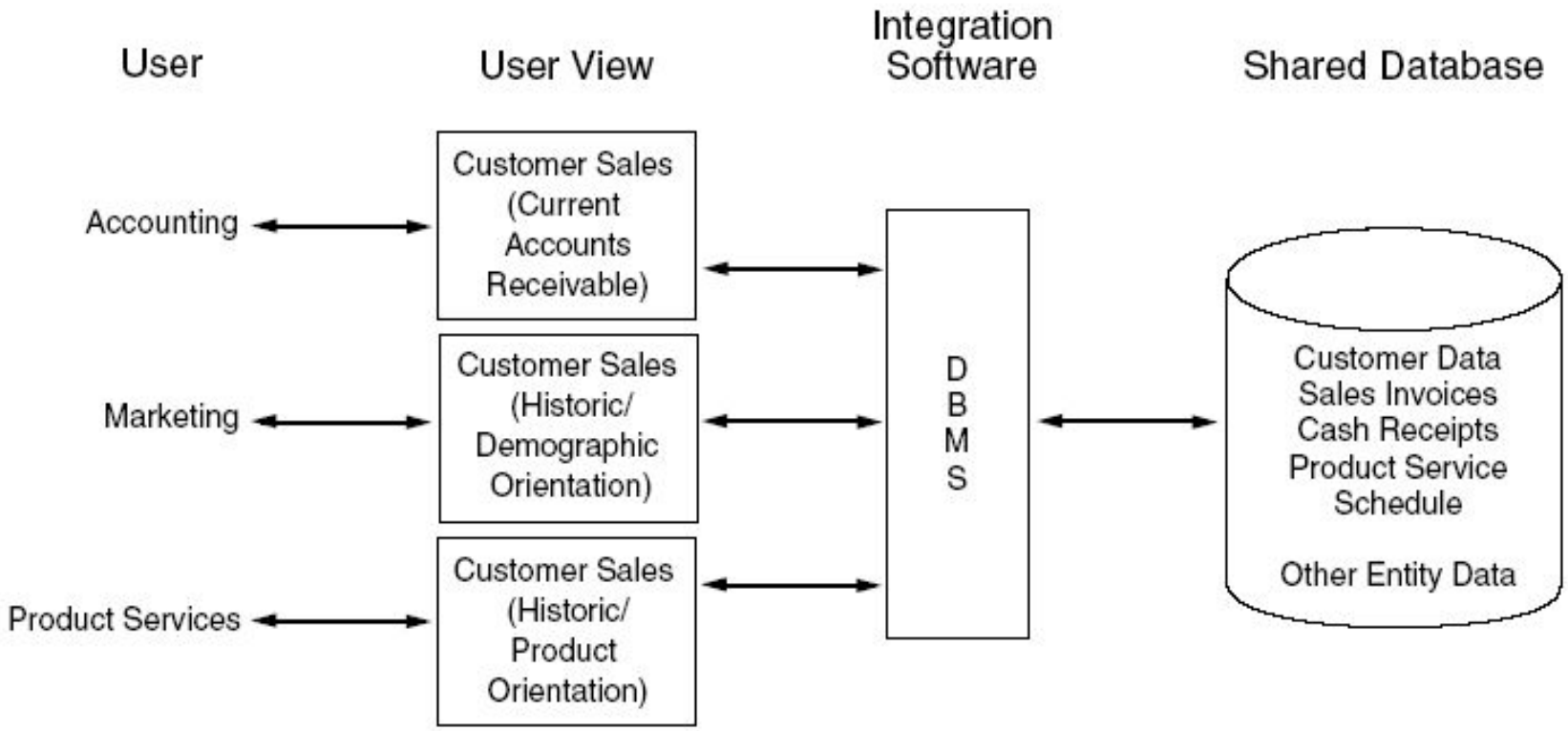
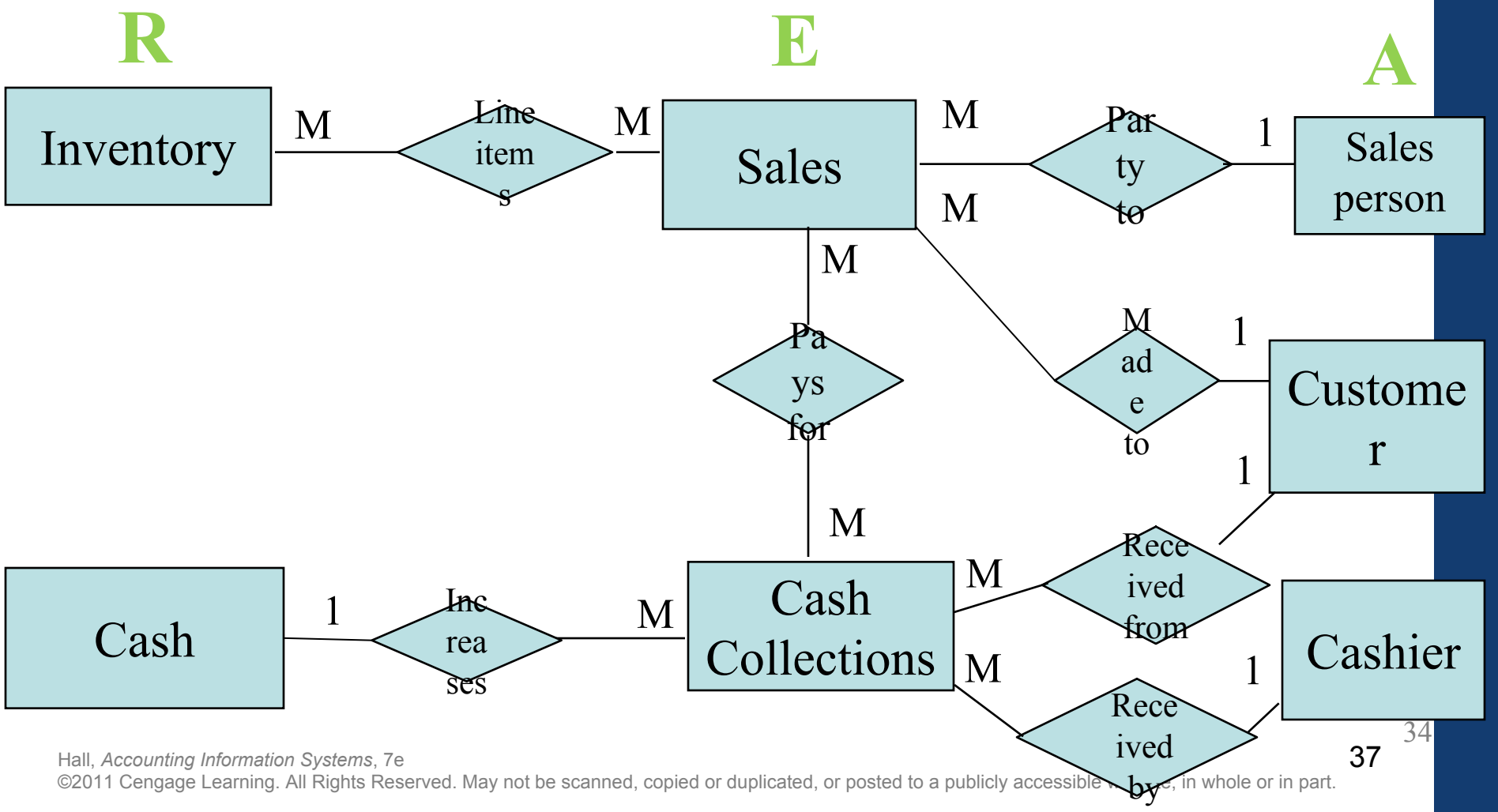


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An REA Data Model Example



Hall, *Accounting Information Systems*, 7e
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REA Model

- The REA model is **an accounting framework for modeling** an organization's
  - economic **resources**; e.g., assets
  - economic **events**; i.e., affect changes in resources
  - economic **agents**; i.e., individuals and departments that participate in an economic event
  - **Interrelationships** among resources, events and agents
- **Entity-relationship diagrams (ERD)** are often used to model these relationships.

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Accountants as Information System Users

- Accountants must be able to clearly convey their needs to the systems professionals who design the system.
- The accountant should actively participate in systems development projects to ensure appropriate systems design.

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Accountants as System Designers

- The accounting function is responsible for the **conceptual system**, while the computer function is responsible for the physical system.
- The **conceptual system** determines the nature of the information required, its sources, its destination, and the accounting rules that must be applied.

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Accountants as System Auditors

- External Auditors
  - attest to fairness of financial statements
  - assurance service: broader in scope than traditional attestation audit
- IT Auditors
  - evaluate IT, often as part of external audit
- Internal Auditors
  - in-house IS and IT appraisal services

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